Course Name:	Web Development Laboratory (316U01L306)	**Semester:**	**III**
Date of Performance:	**4/08/2026**	**DIV/ Batch No:**	**A2-50**
Student Name:	**Kehan Desai**	**Roll No:**	**16010125050**

Experiment No: 2

Title: Design web Portal using CSS3.

Aim of the Experiment: To apply CSS3 style sheets to an existing HTML5-based Web Portal for the given Use Case study in order to enhance layout, responsiveness, visual aesthetics, and user interaction using modern CSS3 features.

Objectives for the Experiment:

1. Implement different CSS3 styling techniques
2. Design responsive layouts using Flexbox and media queries
3. Enhance user interaction using animations
4. Maintain separation of content and presentation

COs to be achieved:

CO1: Use CSS to prepare the layout of web pages.

Books/ Journals/ Websites references:

1. Students should write ……..

Theory:

Students should write about CSS3 (citation needed for content taken (IEEE or APA7 style))

Problem statement: Demonstrate the implementation of CSS3 features to enhance the layout, responsiveness, and aesthetics of the selected HTML-based web portal for the selected Use Case in experiment 1.

Note: All tasks are compulsory to use.

Task 1: Style the Homepage Header (Inline CSS)

Requirements:

- Use **inline CSS** to:
  - Set background color for header
  - Change text color of store name
  - Center align slogan
  - Add padding and border
- Apply inline style to:
  - <h1>, <p>, <header>

Concepts tested:
Inline CSS, color, background, padding, text-align, border.

Task 2: Internal CSS for Navigation Menu

Navigation menu should look professional and consistent across the page.

Requirements:

- Use **internal CSS (<style> in <head>)**
- Style navigation menu:
  - Remove bullet points
  - Display menu items horizontally
  - Add hover effect
  - Change link colors
- Use:
  - :hover, text-decoration, display: inline-block

Concepts tested:
Internal CSS, selectors, pseudo-class, styling links.

Task 3: External CSS for Whole Website Theme

For a consistent look across all pages.

Requirements:

- Create file: style.css
- Link it using <link>
- Apply:
  - Body font-family

- Background color
- Common heading color
- Common footer style

Concepts tested:
External CSS, reusable styles, site-wide theming.

Deliverables : style.css linked with book.html

Task 4: Style Book Categories Sidebar

The categories section should look like a card.

Requirements:

- Use **external or internal CSS**
- Apply:
  - Border
  - Box shadow
  - Padding
  - Rounded corners
- Style list items with:
  - Spacing
  - Hover color change

Concepts tested:
Box model, border-radius, box-shadow, list styling.

Task 5: Book Card Layout Using CSS Box Model

To make Each item appear as a product card.

Requirements:

- Style a as a book card:
  - Width, margin, padding, border
  - Center align content
- Use:
  - display: inline-block
  - Add background color and shadow

Concepts tested:
Box model, margin, padding, display property.

Deliverable : link this .css file with book.html

Task 6: Image Gallery Styling with CSS

To make Featured appropriate items/ attributes, whichever applicable in the Use Case look like a grid.

Requirements:

- Use **CSS** to:
  - Set same size for all images
  - Add border and hover zoom effect
  - Add spacing between images
- Use:
  - transition, transform: scale()

Concepts tested:
Image styling, hover effects, transitions.

Deliverable nav.css , link to nav bar of expt1

Task 7: Table Styling for Price List

Requirements:

- Use CSS to:
  - Collapse borders
  - Add header background color
  - Zebra-striping rows
  - Center align text
- Use:
  - border-collapse, nth-child(even)

Concepts tested:
Table styling, pseudo-classes.

Deliverable table.css , link to the corresponding file.

Task 8: Style Registration Form

Requirements:

- Style:
  - Input boxes width, padding
  - Submit button color and hover effect
  - Fieldset and legend
- Use:
  - :focus, :hover
  - border-radius

Concepts tested:
Form UI styling, pseudo-classes, layout.

Deliverable : form.css

Task 9: Page Layout Using CSS (Header, Sidebar, Content, Footer)

To Create a proper website layout.

Requirements:

- Use CSS to:
  - Set sidebar to left
  - Content to right
  - Footer at bottom
- Use:
  - float or flex
  - Set widths and spacing

Concepts tested:
Layout design, positioning, float/flex.

Deliverable : layout.css

Task 10: CSS Effects & Highlighting.

To attract attention for Special events/data/section, etc..

Requirements:

- Create an “Offer” box:
  - Blinking or glowing effect
  - Different background color
  - Bigger font

- Use:
  - @keyframes animation OR hover effect
  - text-shadow, box-shadow

Concepts tested:
CSS animations, effects, visual emphasis.

Code: (CSS Only)

```
body{
   font-family:Arial,sans-serif;
   margin:0;
   background-color:#f4f9f4;
}

header{
   background-color:#2e8b57;
   color:white;
   text-align:center;
   padding:20px;
}

h2,h3{
   color:#2e8b57;
}

nav{
   background-color:#14532d;
   padding:12px;
   text-align:center;
}

nav a{
   color:white;
   text-decoration:none;
   margin:0 15px;
   font-weight:bold;
}

nav a:hover{
```

```
    color:yellow;
}

section,
article,
aside{
    background-color:white;
    margin:20px;
    padding:20px;
    border-radius:10px;
}

ol,
ul{
    background-color:#f0fff0;
    padding:15px 40px;
    border-left:5px solid #2e8b57;
}

dl{
    background-color:#f8fff8;
    padding:15px;
    border:1px solid #90c695;
    border-radius:5px;
}

dt{
    margin-top:10px;
    color:#14532d;
    font-weight:bold;
}

mark{
    background-color:yellow;
    padding:3px;
}

hr{
    border:1px solid #90c695;
}
```

```
em{
   color:#1b5e20;
   font-weight:bold;
}

small{
   color:gray;
}

aside{
   background-color:#e8f5e9;
   border-left:6px solid #2e8b57;
   box-shadow:0 4px 8px rgba(0,0,0,0.1);
}

aside h3{
   color:#14532d;
   margin-top:0;
}

aside ul{
   list-style:none;
   padding-left:0;
   background:none;
   border:none;
}

aside li{
   padding:10px;
   margin:8px 0;
   background:white;
   border-radius:5px;
   border-left:4px solid #43a047;
}

aside li:hover{
   background:#c8e6c9;
}

.gallery{
   display:flex;
```

```
    justify-content:center;
    gap:20px;
    flex-wrap:wrap;
    margin:20px;
}

.gallery figure{
    background:white;
    padding:15px;
    border-radius:10px;
    border:2px solid #90c695;
    box-shadow:0 4px 8px rgba(0,0,0,0.2);
    text-align:center;
}

.gallery img{
    width:250px;
    height:180px;
    object-fit:cover;
    border-radius:8px;
    transition:transform 0.3s;
}

.gallery img:hover{
    transform:scale(1.05);
}

figcaption{
    margin-top:10px;
    font-weight:bold;
    color:#14532d;
}
.farm-map{
    text-align:center;
    margin:20px;
}

.farm-layout{
    width:600px;
    height:400px;
    margin:auto;
```

```
    border:3px solid #2e8b57;
    display:flex;
    border-radius:10px;
    overflow:hidden;
}

.plot{
    width:33.33%;
    display:flex;
    justify-content:center;
    align-items:center;
    color:white;
    text-decoration:none;
    font-size:25px;
    font-weight:bold;
}

.plot-a{
    background:#66bb6a;
}

.plot-b{
    background:#43a047;
}

.plot-c{
    background:#2e7d32;
}

.plot:hover{
    opacity:0.8;
}

#plot-a,
#plot-b,
#plot-c{
    background:#f8fff8;
    padding:15px;
    margin:15px 20px;
    border-left:5px solid #2e8b57;
    border-radius:5px;
```

```
}

table{
    width:90%;
    margin:30px auto;
    border-collapse:separate;
    border-spacing:0;
    background:white;
    border-radius:15px;
    overflow:hidden;
    box-shadow:0 5px 15px rgba(0,0,0,0.2);
}

caption{
    font-size:26px;
    font-weight:bold;
    color:#14532d;
    margin-bottom:15px;
}

thead th{
    background:#2e8b57;
    color:white;
    padding:15px;
}

tbody td{
    padding:14px;
    text-align:center;
}

tbody tr:nth-child(odd){
    background:#f1f8e9;
}

tbody tr:nth-child(even){
    background:white;
}

tbody tr:hover{
    background:#b9f6ca;
```

```
}

tfoot td{
    background:#14532d;
    color:white;
    padding:15px;
    font-size:18px;
}

th:first-child{
    border-top-left-radius:15px;
}

th:last-child{
    border-top-right-radius:15px;
}

td{
    border-right:1px solid #dcedc8;
}

td:last-child{
    border-right:none;
}

form{
    width:60%;
    margin:20px auto;
}

fieldset{
    border:3px solid #2e8b57;
    border-radius:12px;
    padding:20px;
    background:#f8fff8;
}

legend{
    font-size:20px;
    font-weight:bold;
    color:#14532d;
```

```
}

input[type="text"],
input[type="tel"],
input[type="number"]{
    width:95%;
    padding:12px;
    border:2px solid #90c695;
    border-radius:8px;
}

input[type="text"]:focus,
input[type="tel"]:focus,
input[type="number"]:focus{
    border-color:#2e8b57;
    outline:none;
    box-shadow:0 0 8px rgba(46,139,87,0.5);
}

input[type="submit"]{
    background:#2e8b57;
    color:white;
    border:none;
    padding:12px 25px;
    border-radius:8px;
    cursor:pointer;
    font-size:16px;
}

input[type="submit"]:hover{
    background:#14532d;
}

.main-layout{
    display:flex;
    gap:20px;
    margin:20px;
}

.sidebar{
    width:25%;
```

```
}

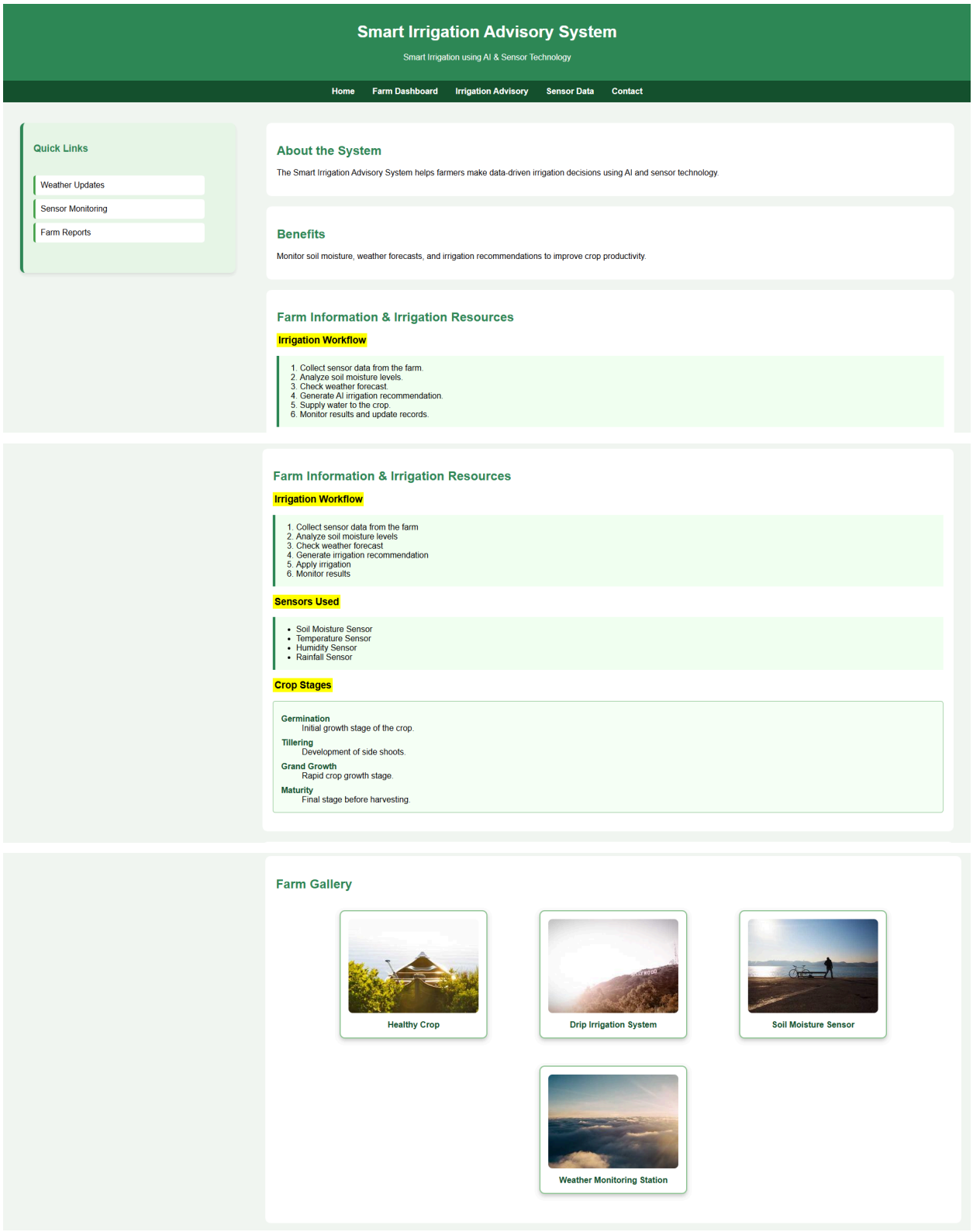
.content{
   width:75%;
}

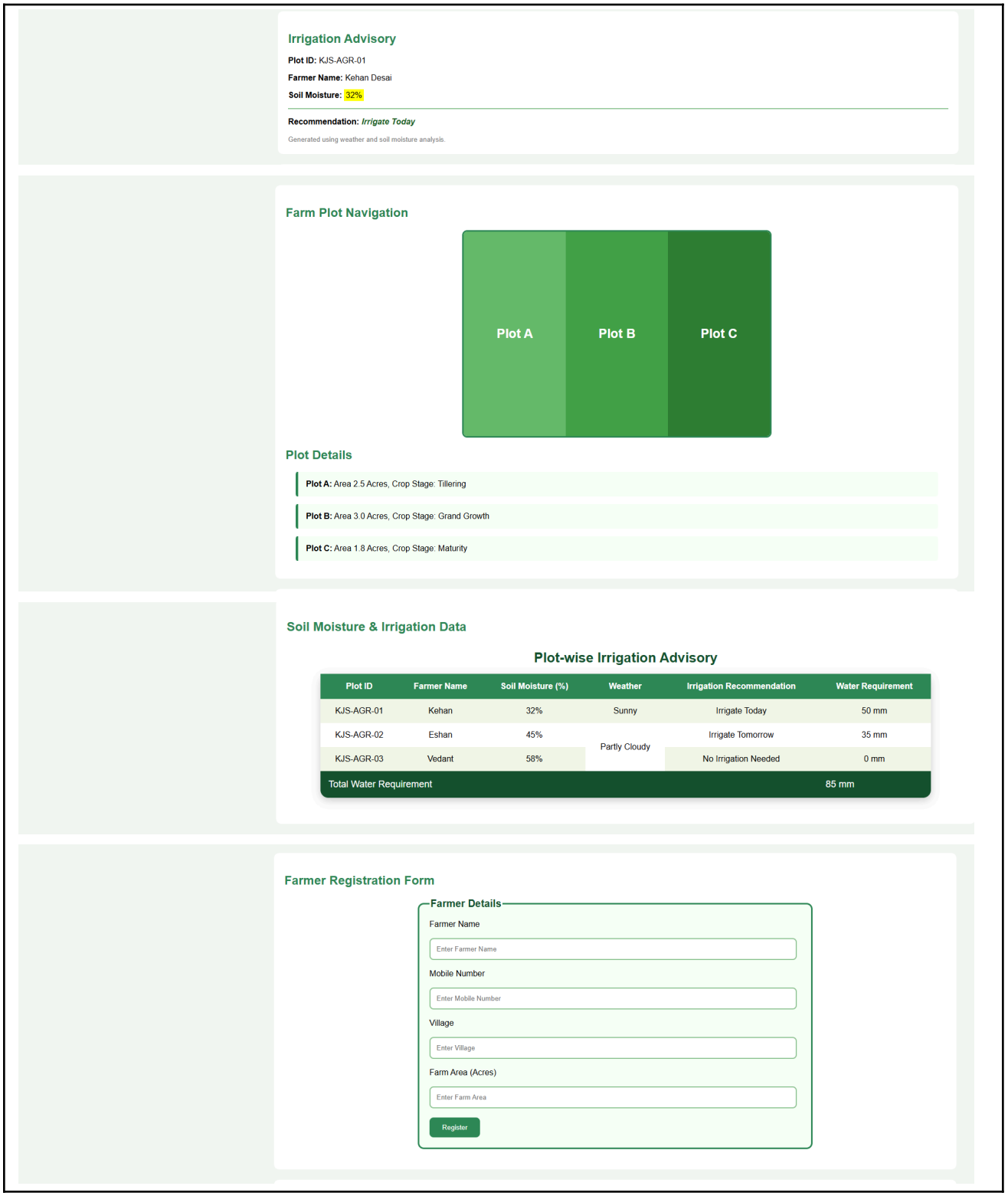
.offer-box{
   width:70%;
   margin:30px auto;
   padding:25px;
   text-align:center;
   background:gold;
   color:#14532d;
   font-size:28px;
   font-weight:bold;
   border-radius:15px;
   box-shadow:0 0 20px orange;
   text-shadow:2px 2px 4px white;
   animation:glow 1.5s infinite alternate;
}

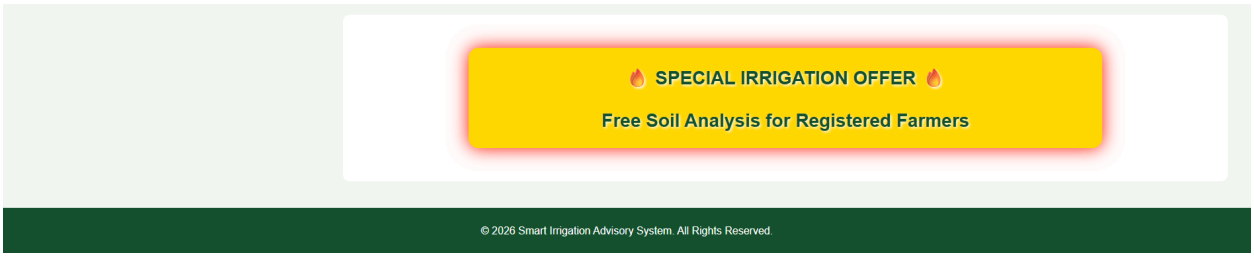
@keyframes glow{
   from{
      box-shadow:0 0 10px orange;
   }
   to{
      box-shadow:0 0 30px red;
   }
}

footer{
   clear:both;
   background:#14532d;
   color:white;
   text-align:center;
   padding:10px;
}
```

Output / Screenshot:




Irrigation Advisory
Plot ID: KJS-AGR-01
Farmer Name: Kehan Desai
Soil Moisture: 32%
Recommendation: Irrigate Today
Generated using weather and soil moisture analysis.
Farm Plot Navigation
Plot A
Plot B
Plot C
Plot Details
Plot A: Area 2.5 Acres, Crop Stage: Tillering
Plot B: Area 3.0 Acres, Crop Stage: Grand Growth
Plot C: Area 1.8 Acres, Crop Stage: Maturity
Soil Moisture & Irrigation Data
Plot-wise Irrigation Advisory
Plot ID
Farmer Name
Soil Moisture (%)
Weather
Irrigation Recommendation
Water Requirement
KJS-AGR-01
Kehan
32%
Sunny
Irrigate Today
50 mm
KJS-AGR-02
Eshan
45%
Partly Cloudy
Irrigate Tomorrow
35 mm
KJS-AGR-03
Vedant
58%
No Irrigation Needed
0 mm
Total Water Requirement
85 mm
Farmer Registration Form
Farmer Details
Farmer Name
Enter Farmer Name
Mobile Number
Enter Mobile Number
Village
Enter Village
Farm Area (Acres)
Enter Farm Area
Register



Post Lab Subjective/Objective type Questions:

1. Use Bootstrap for CSS.

Bootstrap was used to style the Smart Irrigation Advisory System. It provides ready-made CSS classes for layouts, tables, forms, buttons, and navigation, making the website responsive, attractive, and easier to develop with less custom CSS code.

Conclusion and Discussion:

This experiment helped me understand how CSS is used to improve the appearance and usability of a web page. I learned to apply different CSS properties for styling text, images, tables, forms, layouts, and animations. I also understood the use of pseudo-classes, Flexbox, and responsive design techniques. Overall, the experiment enhanced my skills in designing visually appealing and user-friendly web pages.

Feedback from Classmates:

1. The website layout is neat and easy to navigate.
2. The form and table styling improve readability.
3. The image gallery and offer box make the webpage more attractive.

Faculty Feedback:

The experiment was implemented correctly and demonstrates a good understanding of CSS styling, layouts, and visual effects.